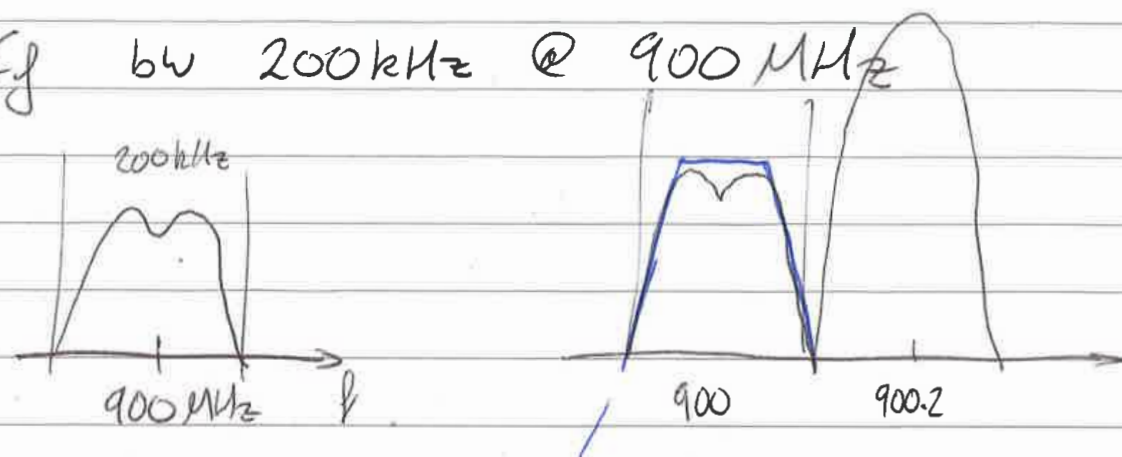


Filters

Adjacent channel

1. Xf bw 200kHz @ 900MHz



Filter response

So reduce by 40dB

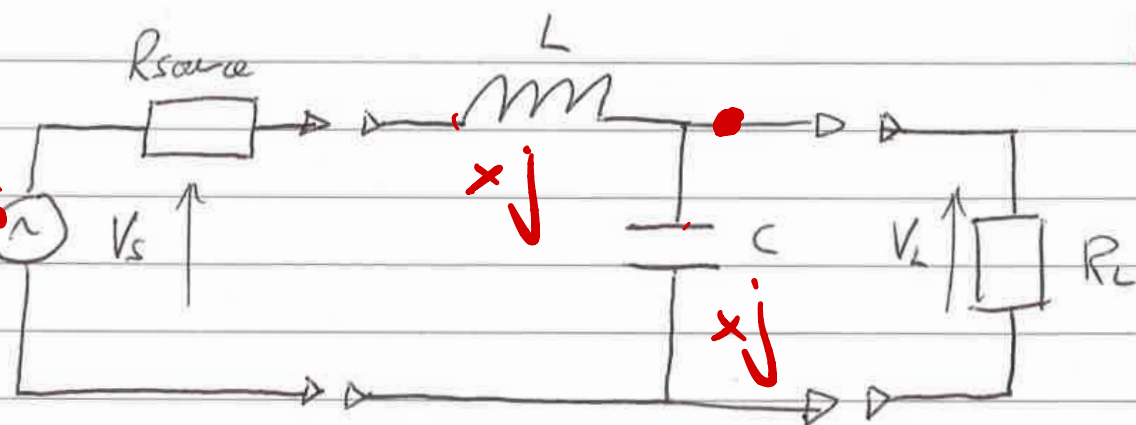
Signal Interferer may be 25dB higher than wanted sig.

Signal must have 15dB over interferer

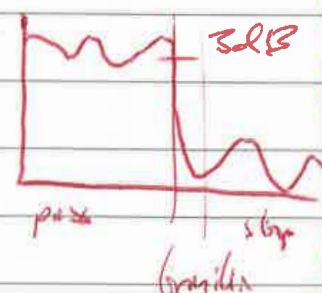
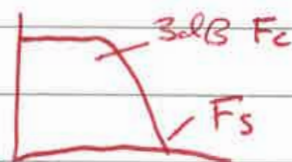
So 40dB

@ dc

$$V_L = \frac{R_L}{R_L + R_S} V_S$$



$$R_L = 1\Omega \quad R_S = 0.1\Omega$$



s-plane transfer functions

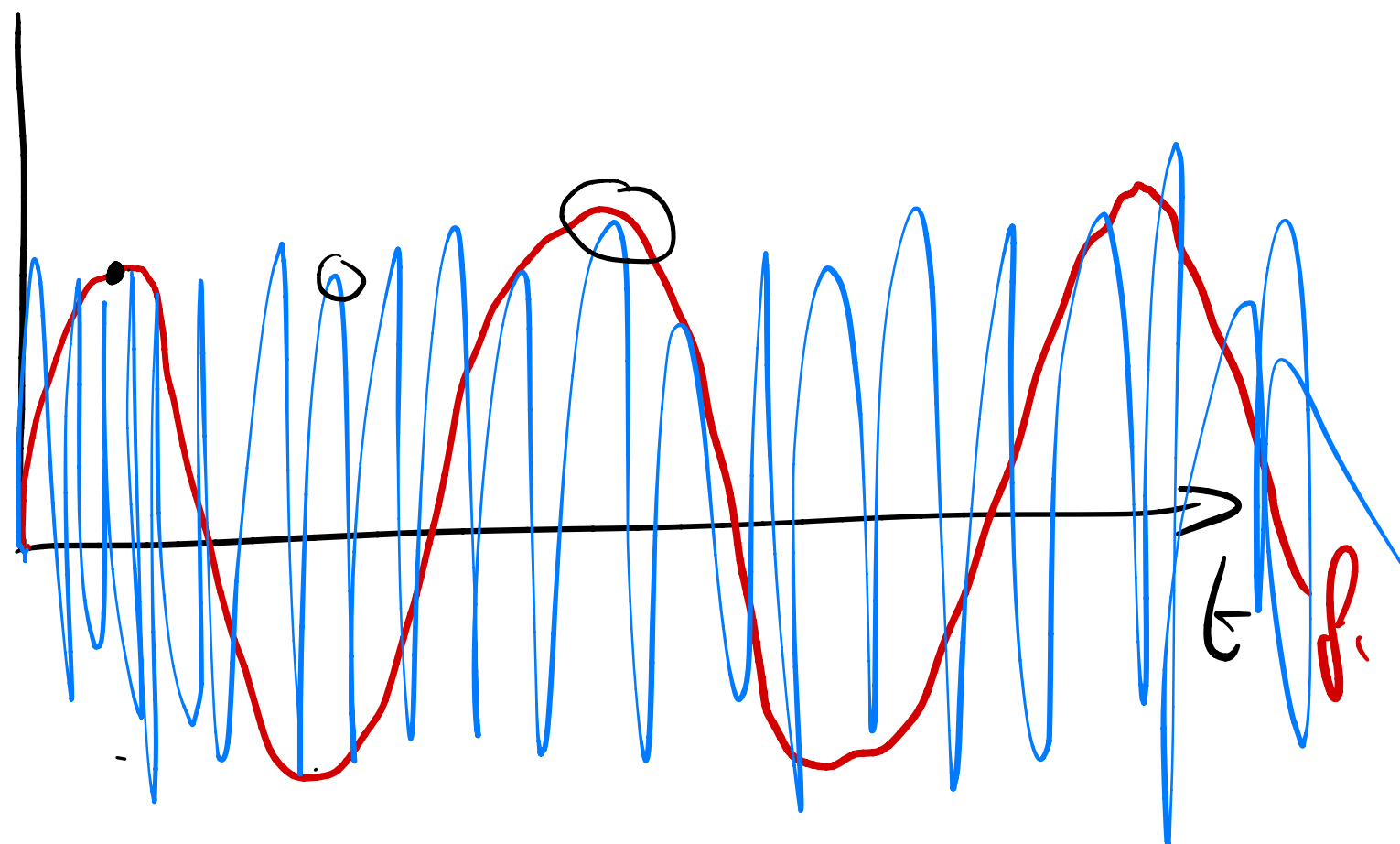
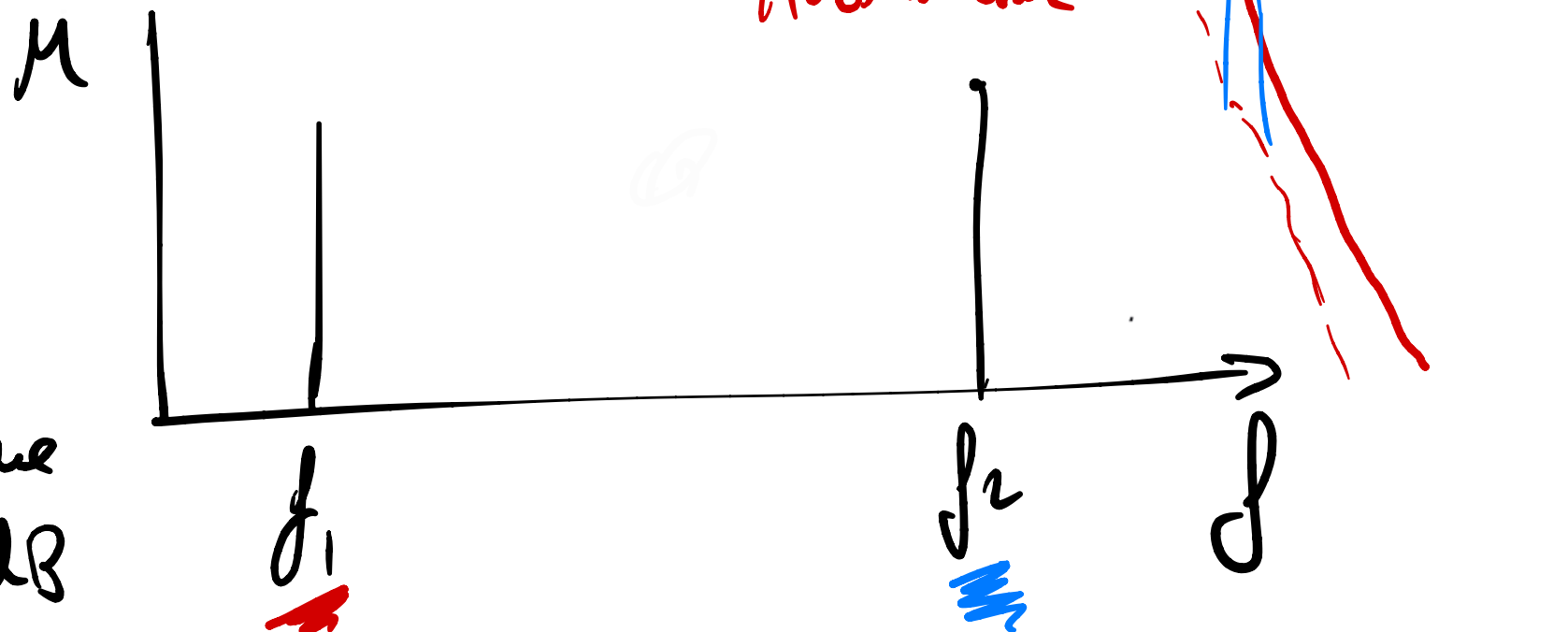
$$R_L \parallel C = \frac{R_L \frac{1}{sC}}{R_L + \frac{1}{sC}}$$

$$\Rightarrow \frac{R_L \frac{1}{sC}}{R_L + \frac{1}{sC}} \Rightarrow \frac{R_L \frac{1}{sC}}{R_L + \frac{1}{sC}} \Rightarrow \frac{1}{1 + sC}$$

$$H(s) = \frac{1}{1 + sC} \quad \frac{1}{1 + sC} + sL + 0.1$$

Filter $\Rightarrow$

Abberation



M

Pass Band

Stop Band

Low pass filter.

$$= \frac{1}{1+sL(1+sC) + 0.1(1+sC)} = \frac{1}{s^2 LC + s(L+0.1C) + 1.1}$$

$$H(s) = \frac{1}{LCs^2 + s(L+0.1C) + 1.1}$$

pole = -6dB/octave
-20dB/decade

$$\text{Butterworth response} = \frac{1}{s^2 + \sqrt{2}s + 1}$$

zero = +6dB/octave
+20dB/decade

$$\text{Match DC gain } \frac{1}{1.1} \times \frac{1}{s^2 + \sqrt{2}s + 1} = \frac{1}{1.1s^2 + 1.1\sqrt{2}s + 1.1}$$

(s+0) pole s=0
bigger or just decay

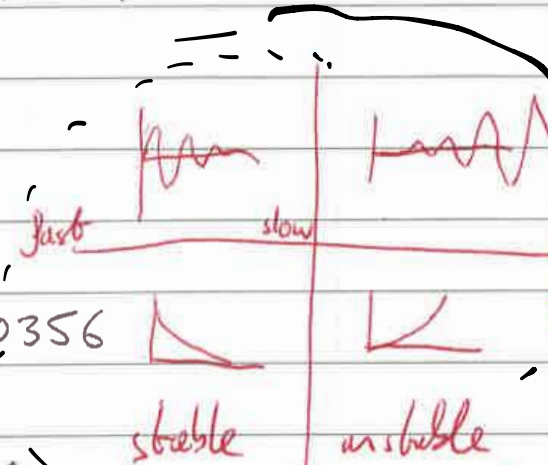
$$\circ \circ LC = 1.1 \text{ e } C = \frac{1.1}{L} \quad L + 0.1C = 1.1\sqrt{2}$$

$$\circ \circ L^2 - 1.1\sqrt{2}L + 0.1 \times 1.1 = 0$$

$$\text{Quadratic } L = 0.7778 \pm 0.70356$$

$$\circ \circ L = 1.481 \text{ or } 0.074$$

$$C = 0.742 \text{ or } 14.82$$



For source 50 and load of 0.025 e corner frequency of 20MHz find values

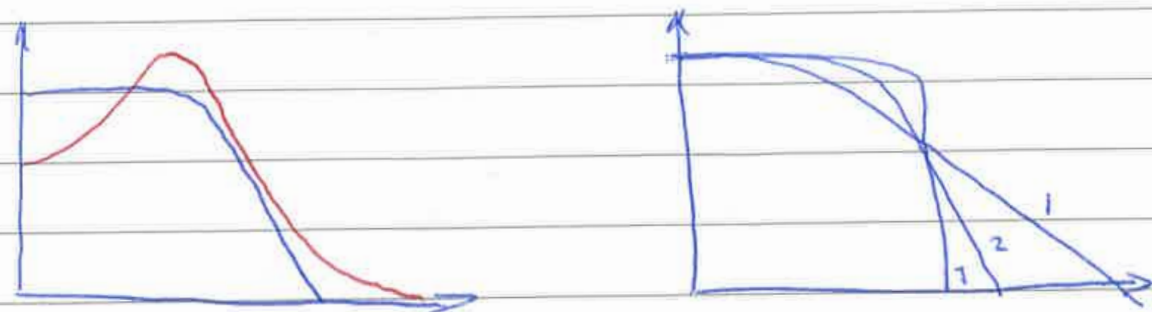
$$\text{Choose } L = 0.074 \Rightarrow \frac{0.074 \times 50}{2\pi \cdot 20 \times 10^6} \text{ e } C = \frac{14.82}{2\pi \cdot 50 \cdot 20 \times 10^6}$$

$$= 29.4 \mu\text{H} \text{ e } 2.36 \text{ nF}$$



Filter Design tables.

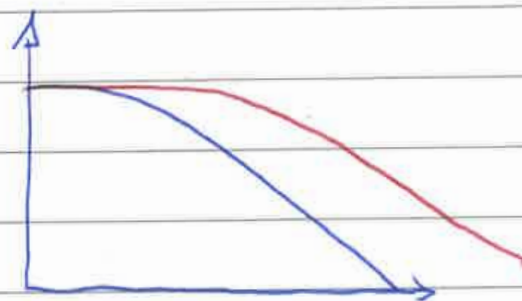
Butter



Flat gain across passband.
Group delay / phase varies.

$$\begin{aligned} 1 & (s+1) \\ 2 & s^2 + \sqrt{2}s + 1 \\ 3 & (s+1)(s^2 + s + 1) \end{aligned}$$

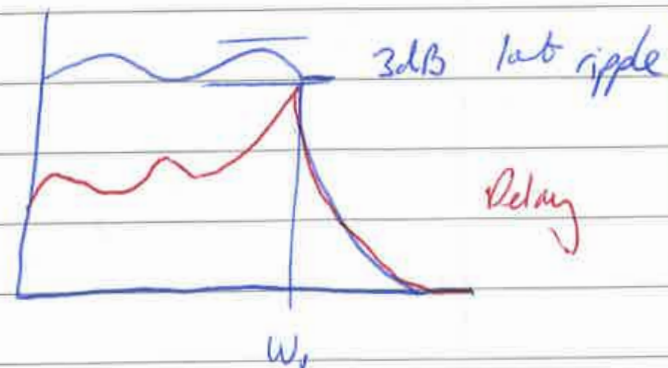
Bessel



Group & phase delay flat across band.

$$\begin{aligned} 1 & s+1 \\ 2 & s^2 + 3s + 3 \\ 3 & s^2 + 6s^2 + 15s + 15 \end{aligned}$$

Chos 1 Passband Ripple } Steeper roll off the Butter
Chos 2 Stopband Ripple } - Not as fast roll off



$$H(s) = \frac{1}{\sqrt{1 + \epsilon^2 T_n^2\left(\frac{\omega}{\omega_0}\right)}}$$

$$\frac{1}{\sqrt{1 + \frac{1}{1 + \epsilon^2 T_n^2\left(\frac{\omega}{\omega_0}\right)}}}$$

Elliptic Filter



$$G_n(\omega) = \frac{1}{\sqrt{1 + \epsilon^2 K_n^2(\frac{\omega}{\omega_0})}}$$

Rewrite $H(s) = \frac{H_0}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$

$$Q = \frac{F_0}{F_H - F_L}$$

20dB - 20dB

Q = more selective

> 0.707

peaking

$\alpha = \frac{1}{Q} \Rightarrow$ Damping Ratio $\xi = 2\alpha$

< 0.707

Filter Table

20dB

$F_c = 4\text{MHz}$ $F_s = 8\text{MHz}$ $A_{\min} = 20\text{dB}$ $Z_0 = 50\Omega$

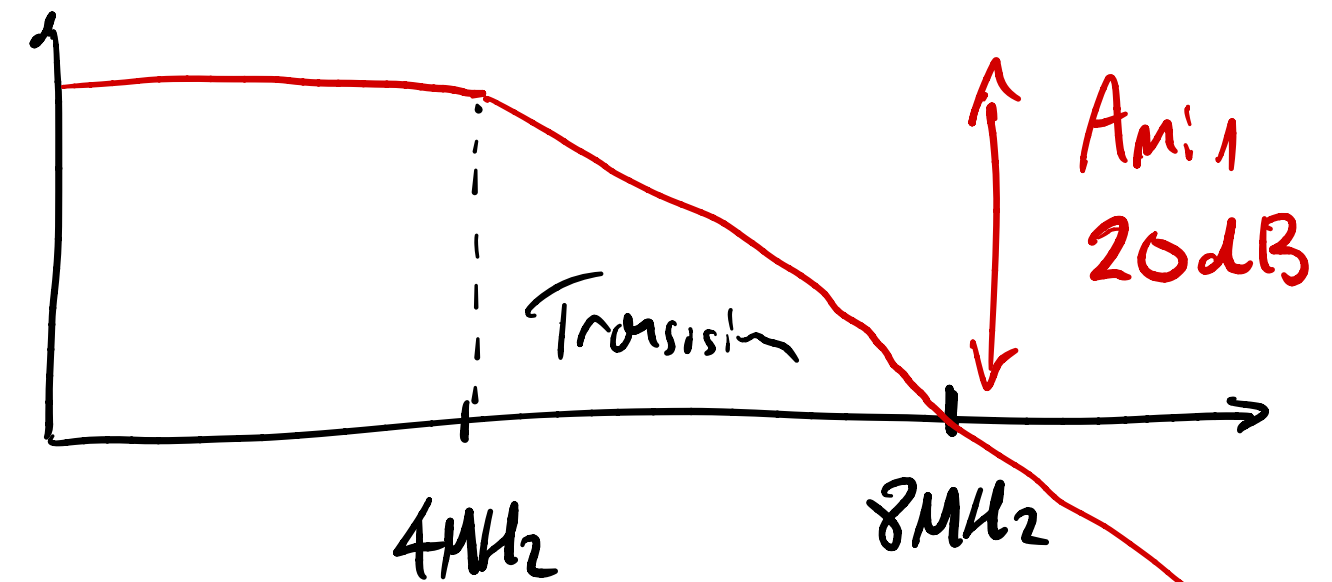
$A_s = \frac{F_s}{F_c} = \frac{8}{4} = 2$ $n = 4$ order

	L_1	C_1	L_2	C_2
4 th	0.7654	1.8478	1.8478	0.7654

Scale =	1.5 μH	1.5 nF	3.7 μH	0.6 nF
---------	-------------------	--------	-------------------	--------

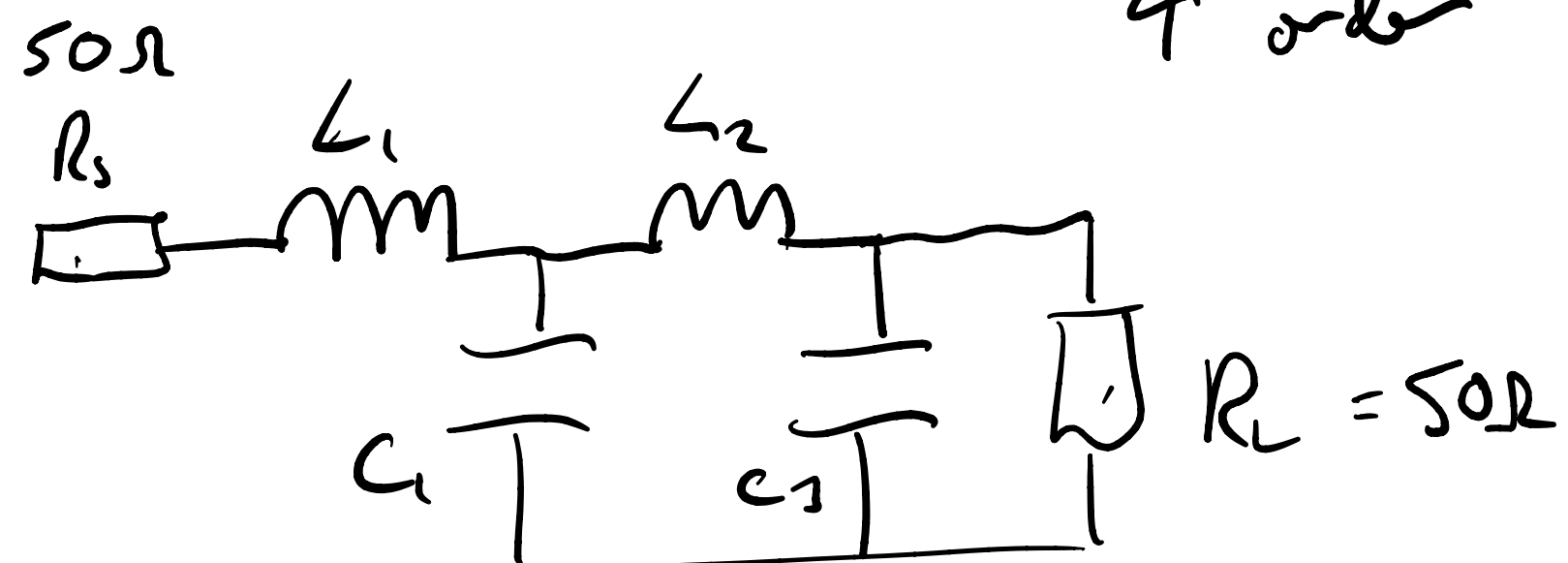
Sketch

Low pass



$$A_s = \frac{F_s}{F_c} = \frac{8}{4} = 2$$

4th order



$\Rightarrow$ Normalized to 1Hz & 1 Ω
Need to scale them

$$L = \frac{L R_0}{2\pi f_c} \quad C = \frac{C}{2\pi f_c R_0} \quad R = R \cdot R_0$$

Lowpass filter synthesis

Passive ladder network realisation

Butterworth

$$A_s = \frac{F_s}{F_c}$$

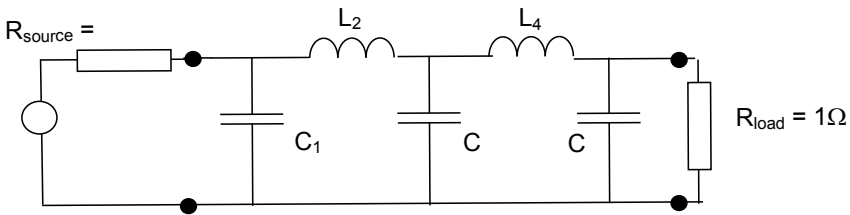
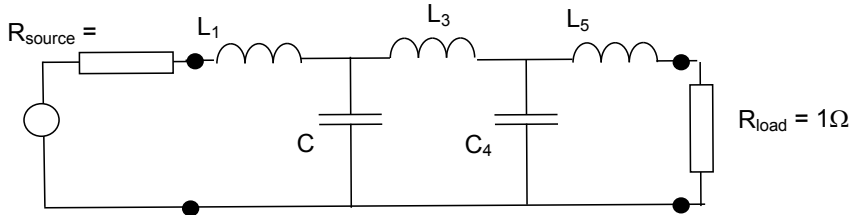


Figure 1. 5th order ladder network dual structures

In order to scale for different values:

$$L \Rightarrow \frac{L.R_o}{2\pi F_c} \qquad C \Rightarrow \frac{C}{2\pi F_c R_o} \qquad R \Rightarrow R.R_o$$

4th



n	C ₁	L ₂	C ₃	L ₄	C ₅	L ₆	C ₇
2	1.4142	1.4142					
3	1	2	1				
4	0.7654	1.8478	1.8478	0.7654			
5	0.618	1.618	2	1.618	0.618		
6	0.5176	1.4142	1.9319	1.9319	1.4142	0.5176	
7	0.445	1.247	1.8019	2	1.8019	1.247	0.445
	L ₁	C ₂	L ₃	C ₄	L ₅	C ₆	L ₇

For example, if a Butterworth filter with $F_c = 3\text{MHz}$ and $F_s = 6\text{MHz}$ with $A_{\text{min}} = 30\text{dB}$ is to operate between 50Ω source and load impedances, then the structure is:



Active VCVS realisation

Buffer acb + filter

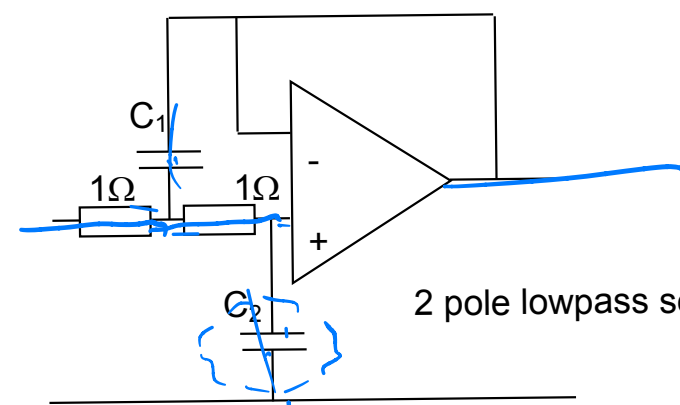
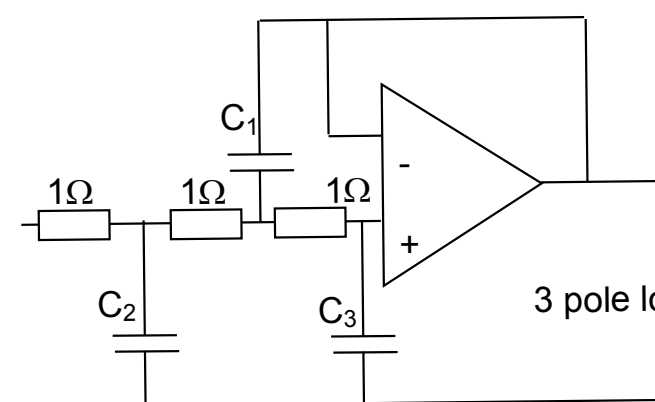


Figure 2. active 2 pole & 3 pole lowpass sections

2 pole lowpass section



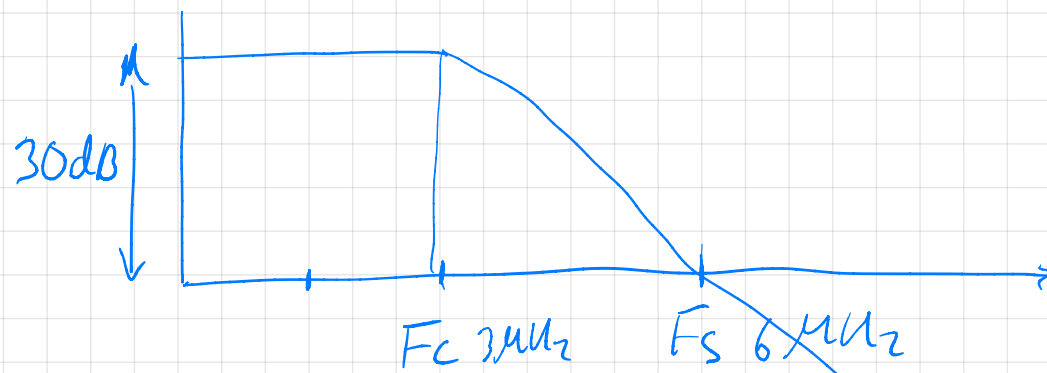
3 pole lowpass section

$$F_c = 3 \text{ MHz}$$

$$F_s = 6 \text{ MHz}$$

$$A_{MV} = 30 \text{ dB}$$

$$R_o = 10 \text{ k}\Omega$$



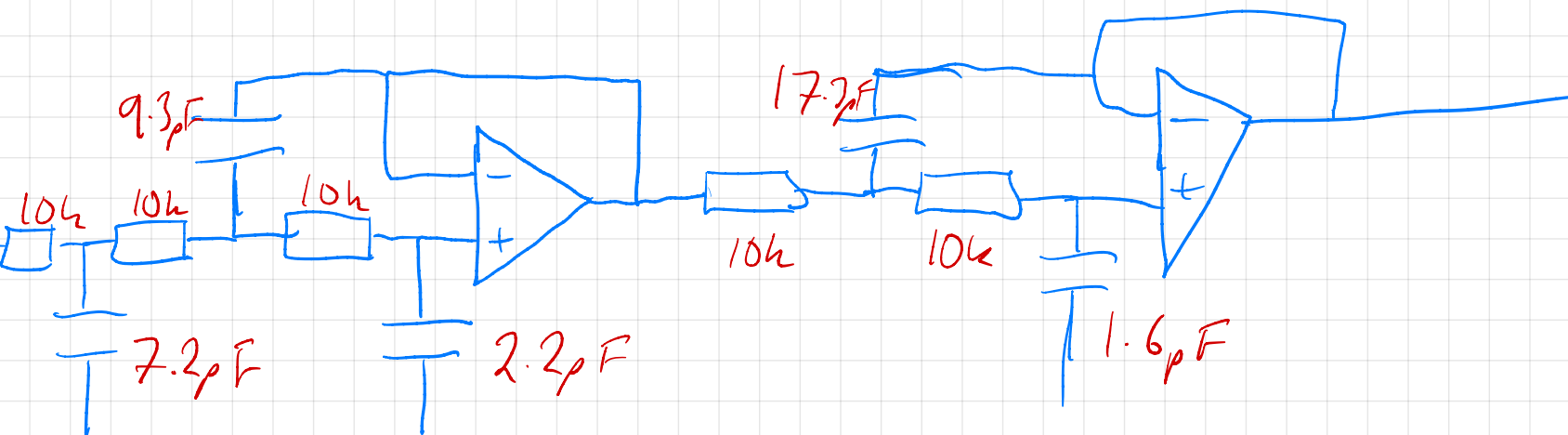
$$A_s = \frac{6}{3} = 2$$

Order = 5 (just)

	C_1	C_2	C_3	Scale
	1.753	1.334	0.4214	$\frac{C}{2\pi f_c R_o}$
	3.251	0.309		
Scale	<u>9.3 pF</u>	7.2 pF	<u>2.2 pF</u>	
	17.3 pF	1.6 pF		

$$10 \times 10^4 = R = \underline{10 \text{ k}\Omega}$$

3rd



There is a wide variety of active structures used in the realisation of filters. Figure 2 shows two pole and three pole VCVS sections. These can be cascaded to realise any order of filter. The operational amplifiers act as buffers and prevent one section from loading another. Once again, normalised element values are available in tables. For the sections shown, all of the resistors are 1Ω and appendix 1 gives the capacitor values for the Butterworth transfer function.

The resistor values can be scaled to a convenient value to make the capacitor values practical. The capacitor values are then scaled in the same way as those in the passive design.

Example 1: Design a Butterworth active lowpass filter with $F_c = 1\text{KHz}$ to provide an attenuation greater than 10dB at 2.5 KHz. Use 10K resistors for the design.

Example 2: Design a Butterworth active lowpass filter with $F_c = 1\text{ KHz}$ and to provide an attenuation greater than 30dB at 2 KHz. Use 10K resistors for the design.

Appendix 1

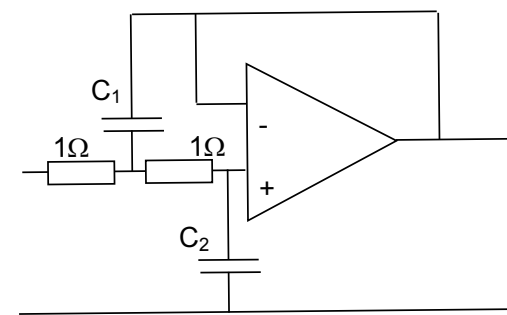
normalised Butterworth active element values

n	C ₁	C ₂	C ₃
2	1.414	0.7071	
3	3.546	1.392	0.2024
4	1.082	0.9241	
	2.613	0.3825	
5	1.753	1.354	0.4214
	3.253	0.309	
6	1.035	0.966	
	1.414	0.7071	
	3.863	0.2588	
7	1.531	1.336	0.4885
	1.604	0.6235	
	4.493	0.2225	

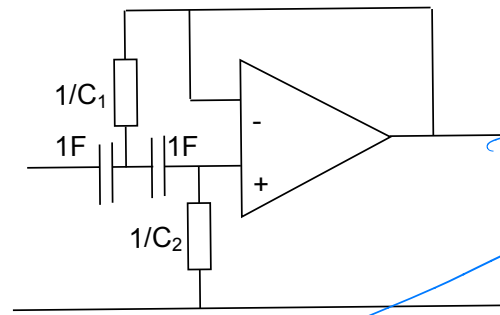
Repeat last example but HIGH pass.

Active Highpass design

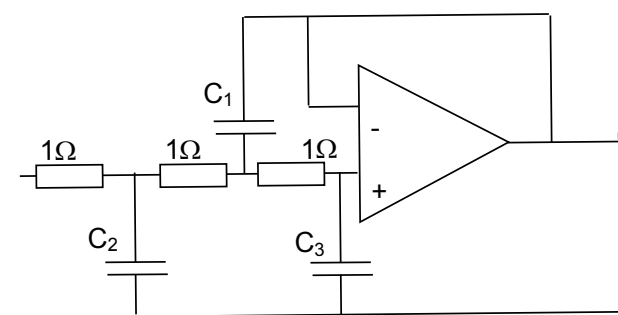
Active highpass filters can be obtained by converting resistors to capacitors and capacitors to resistors in the normalised lowpass sections. The required filter order is obtained using the process already described for passive highpass filters.



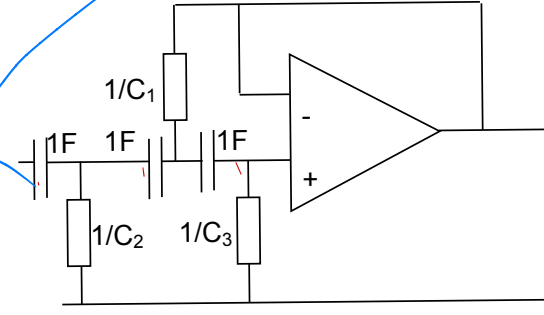
2 pole lowpass section



2 pole highpass section



3 pole lowpass section



3 pole highpass section

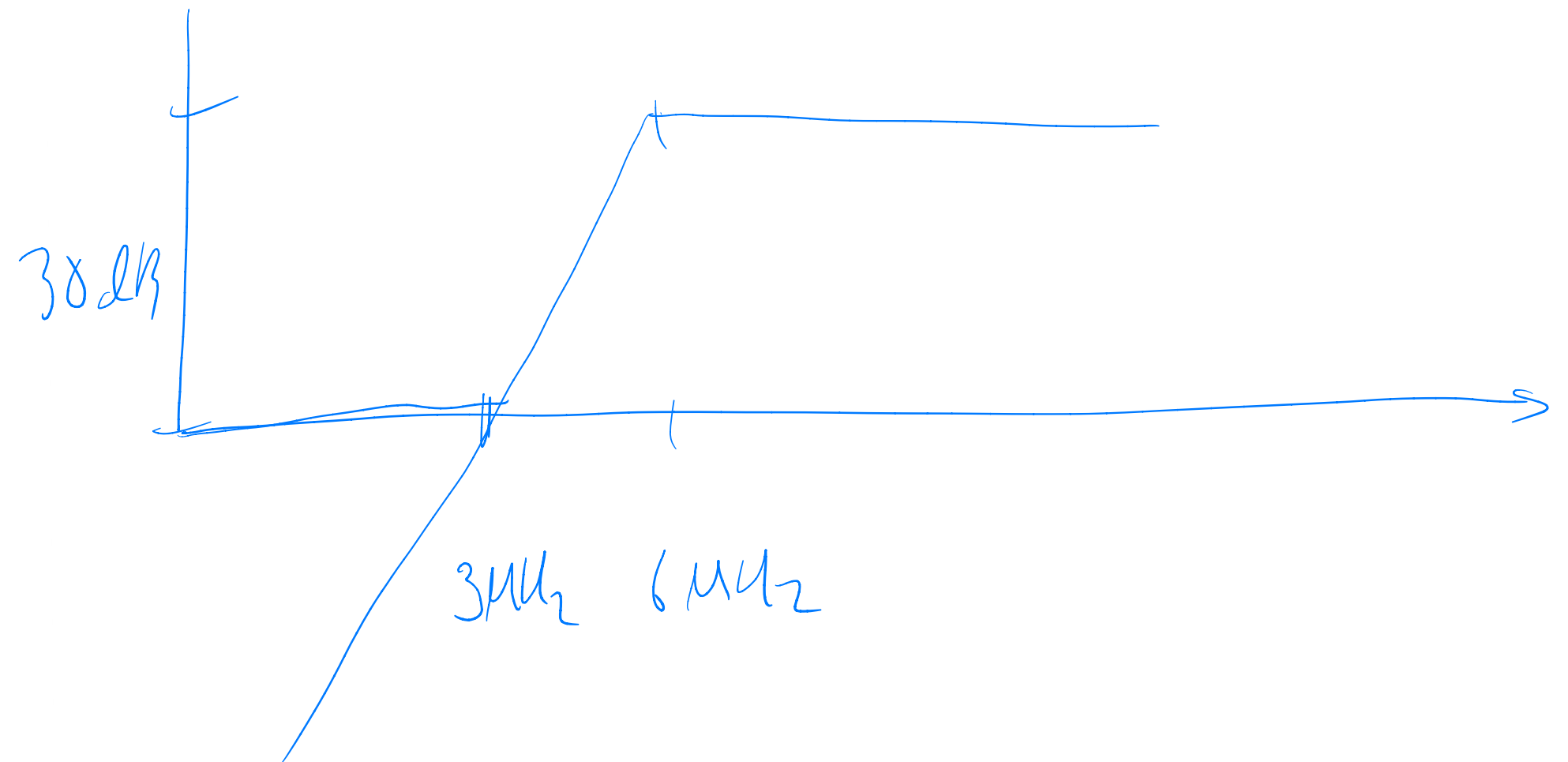
$$R_{HP} = \frac{1}{C_{LP}} \quad \text{and} \quad C_{HP} = \frac{1}{R_{LP}}$$

This transformation only applies to components which are in the frequency selective path, not resistors used to set the amplifier gain (eg the resistors associated with inverting or non-inverting op amp configuration gain). The components are then scaled for corner frequency (F_c) and impedance level (Z):

$$R_{HP} \Rightarrow R_{HP} Z \quad \text{and} \quad C_{HP} \Rightarrow \frac{C_{HP}}{Z 2\pi F_c}$$

The reactance of the capacitors in the first section should be much higher than the source impedance at the highest required passband frequency.

CFH Jan2003



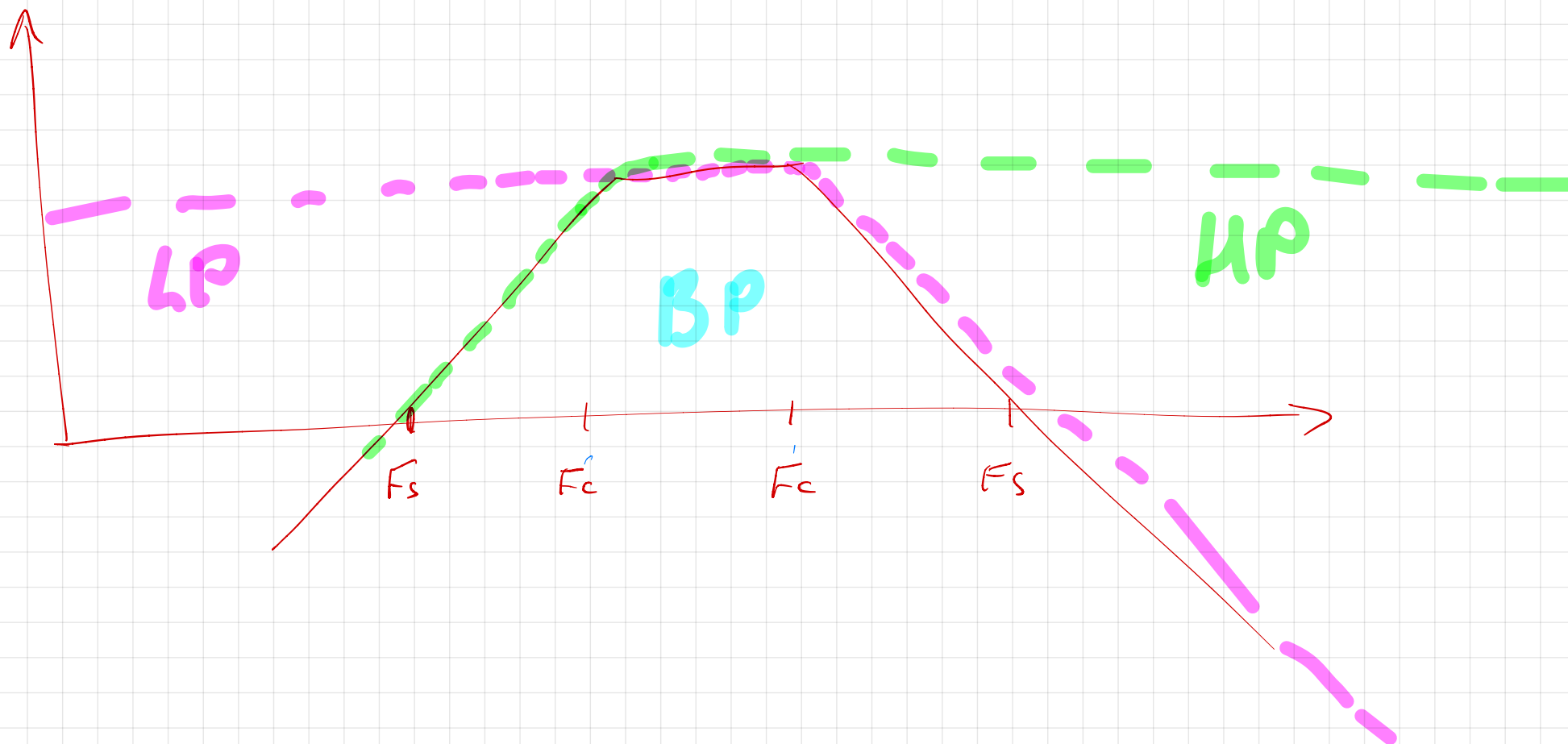
$$LP \quad A_s = \frac{F_s}{F_c}$$

$$HP \quad A_s = \frac{F_c}{F_s} = 2$$

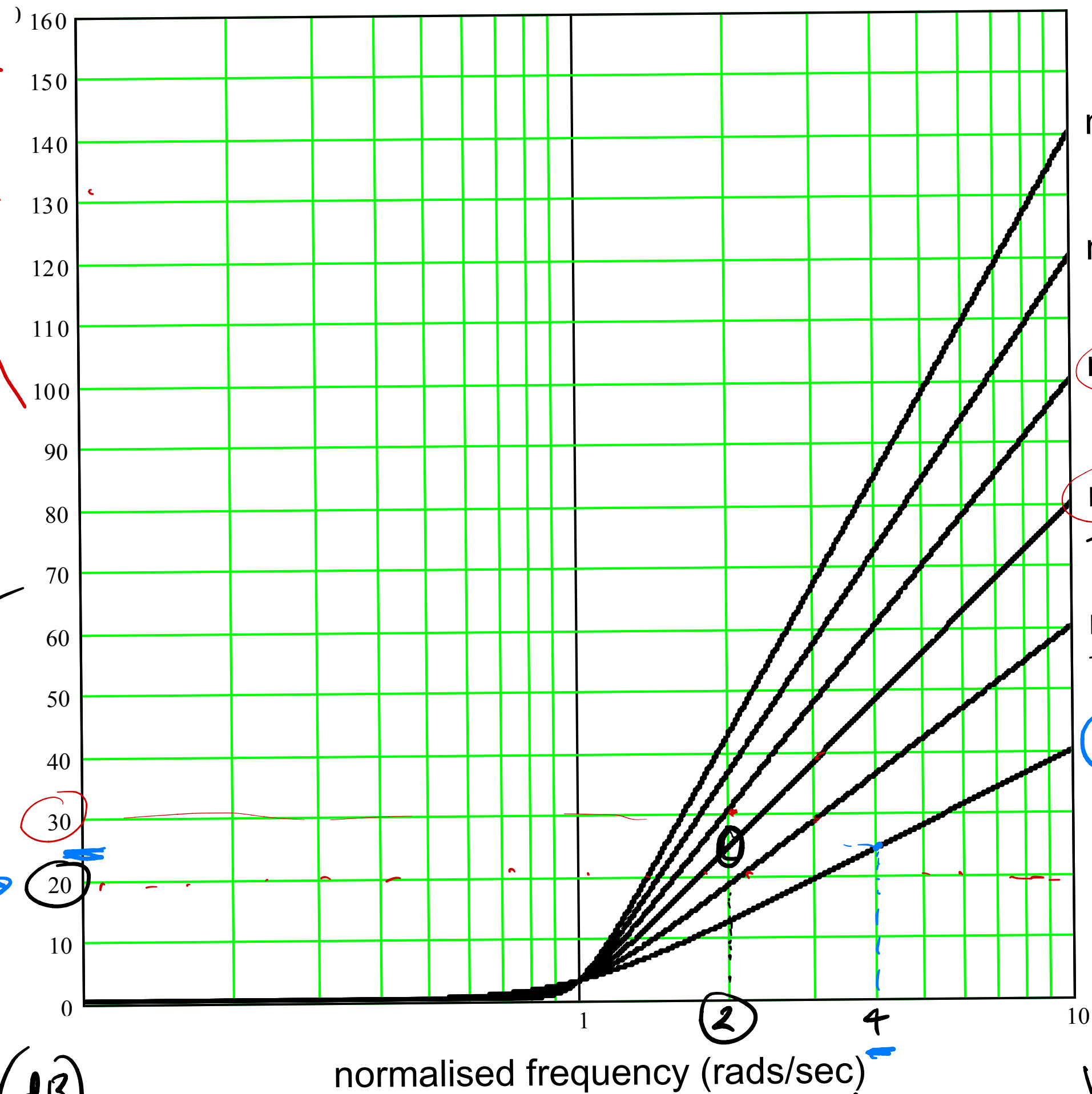
Order = 5th

Start with C $C_{HP} = \frac{1}{R_{LP}} \Rightarrow C = \frac{1}{\frac{1}{1}}$

R_1	R_2	R_3	} $\times 10^4 \Rightarrow$				scale $\frac{C}{2\pi f_c R} = \underline{\underline{2.7pF}}$
$\frac{1}{1.751}$	$\frac{1}{1.354}$	$\frac{1}{0.4214}$		5.7k	7.4k	23.7k	
$\frac{1}{3.253}$	$\frac{1}{0.109}$			3k	32.4k		



Butterworth attenuation (dB) n=2 to n=7



Highpass, bandpass & bandstop passive filters

The first step in the design of any passive filter is to determine a suitable transfer function and filter order that meets the specified requirements. Once the transfer function type has been established, the same curves that are used to design lowpass filters can be used to determine the filter order.

Once the order is established, the same lowpass tables can be used to obtain the element values for the corresponding normalised lowpass filter. This filter can then be converted to the desired type and scaled to the required impedance and cut-off frequency or centre frequency and bandwidth.

The conversion process is illustrated in figure 1 below:

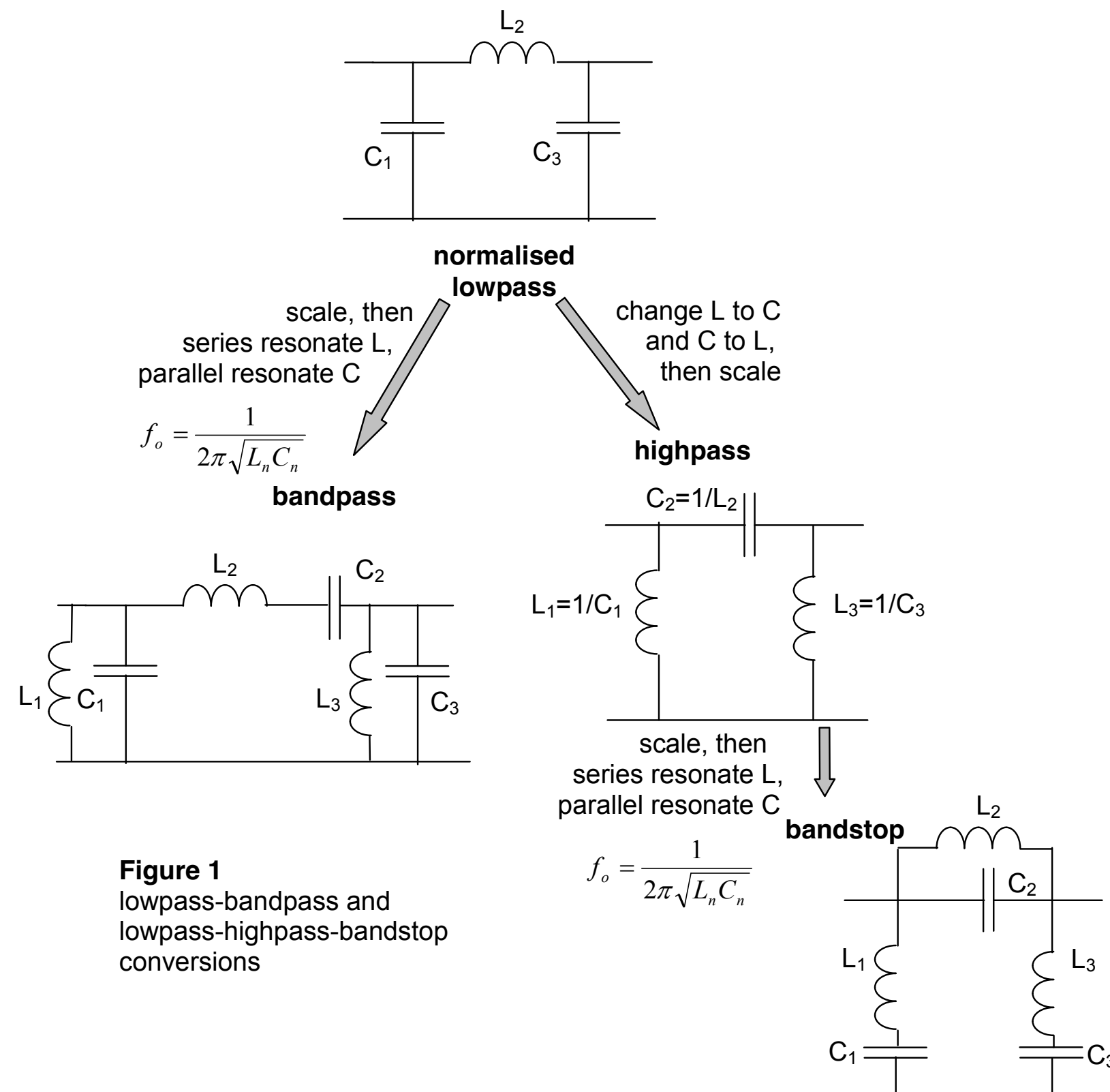
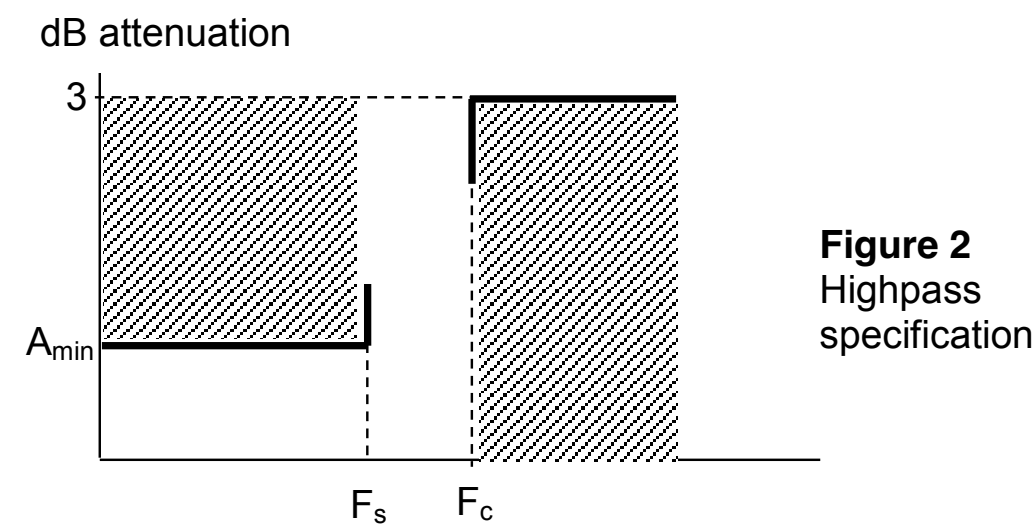


Figure 1
lowpass-bandpass and
lowpass-highpass-bandstop
conversions

Highpass design

Figure 2 shows the specification for a highpass filter. The shape factor for the filter is determined in a similar manner to the lowpass filter but using:

$$A_s = \frac{F_c}{F_s}$$



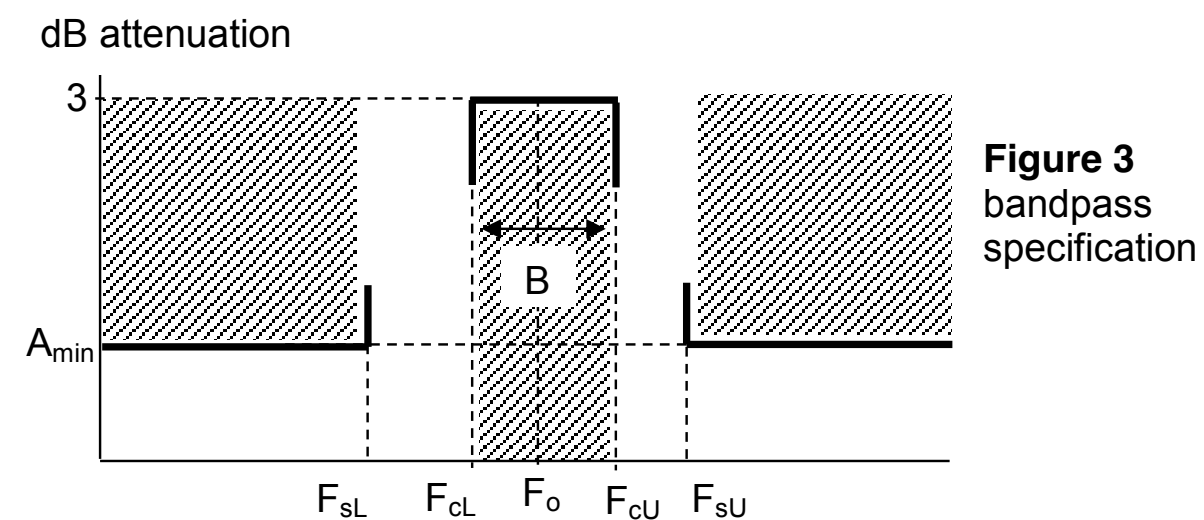
The required order is then found in exactly the same way as the lowpass filter using the same design curves.

The normalised lowpass element values are obtained from tables and the filter is then converted to a normalised highpass filter by replacing all inductors with capacitors and vice versa. The values of the new highpass elements are obtained from the reciprocal of the lowpass values (note: Henries become Farads and vice versa).

The filter is then scaled in exactly the same way as a lowpass filter to achieve the necessary -3 dB frequency and impedance.

Bandpass design

Figure 3 shows the specification for a bandpass filter.



In this case, the specification will include the upper and lower –3dB frequencies together with upper and lower stopband frequencies at which the specified attenuation must be achieved.

The bandpass frequency response of a practical filter follows **geometric symmetry**. In other words, the relationship between the upper and lower –3dB frequencies and the centre frequency is given by:

$$F_o = \sqrt{F_{cL} F_{cU}}$$

The bandwidth is related to the upper and lower –3dB frequencies by:

$$B = F_{cU} - F_{cL}$$

The centre frequency, bandwidth and bandpass Q factor are related by:

$$B = \frac{F_o}{Q}$$

There are a number of different realisation strategies that depend upon the required bandpass Q (ie whether the filter is wide band or narrow band).

If the filter is narrowband (ie bandwidth less than 15% of centre frequency) the a **coupled resonator** architecture usually gives practical realisable component values. The design of these filters will not be discussed here.

If the filter is wideband (ie bandwidth equal to centre frequency, one octave between F_{cL} and F_{cU}) then a **lowpass-highpass cascade** can be used. This consists of a lowpass filter designed using a shape factor F_{sU}/F_{cU} in series with a highpass filter designed using shape factor F_{cL}/F_{sL} . This is illustrated in figure 4.

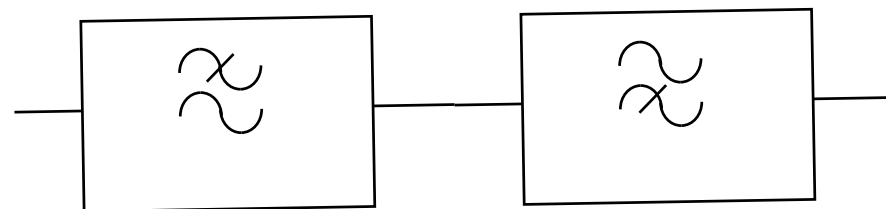


Figure 4
lowpass-highpass cascade bandpass filter